

Appendix B:

We trained a hobby-pianist over 140 hours to memorize the mapping, generate text and transcribe given sentences. The training spanned 25 weeks with on average of 6 hours per week. A large number of exercises was created focusing on different aspects of typing, piano playing, chords and redundancy. The participant trained on her own during the week, following the instructions given by the training software, as shown in Figure 1. Once a week we met with the participant to conduct a controlled performance test. This meeting was also used to give feedback on finger movements and typing manners and to discuss new exercises.

The first three weeks were dedicated to memorize the letter mapping. Chords were introduced in blocks of 4–10 per week, starting within week 7. They accompanied other exercises for gaining speed when typing with single letters. Based on the performance in previous weeks, we created special exercises that focused on statistically slow letter sequences, and increased the complexity of phrases. We introduced practices for large lateral movements (jumps within one hand), the combination of both hands on different parts of the piano and the spontaneous reaction on infrequent letter pairs. It was taken care that basic exercises, like frequent bigrams, chord and single letter mappings, were repeated with an expanding schedule. In addition to the training software, the participant was told to practice her skills in online typing games. Table 1 shows the exercise schedule and example stimuli. Sentences were taken from the Enron email data set [3] and websites for training english as a second language [1, 2].

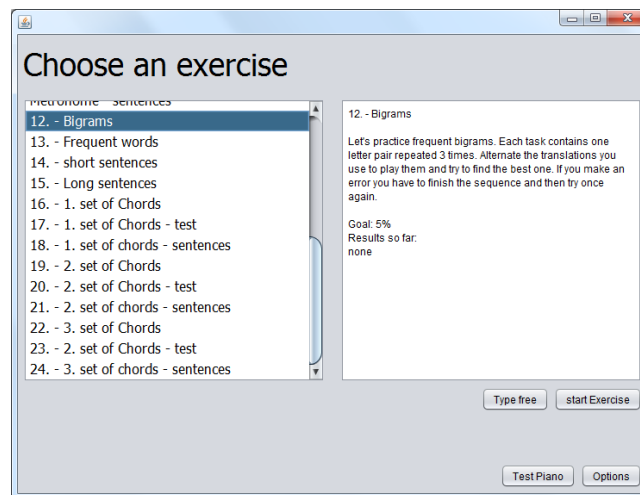


Figure 1: The main screen of the training software. The list on the left displays the available exercises. The right window shows the description of the chosen exercise and previous performances

Weeks	Exercise	Stimulus Example	Goal
1– 3	Letter-to-key mapping	see Figure 2	Memorize letter positions
from 3	Frequent bigrams and words	<i>th in es st an they can give only want</i>	Resolve multiplicity, memorize best translation
	short sentences	<i>if he wants it i am on my way i am on a plane [3]</i>	speed up
7–14	chord-mapping	see Figure 3	memorize chords that enter a sequence of letters
from 7	sentences containing n-grams mapped to chords	<i>i am not aware of any joel will handle this what a nice note [3]</i>	detect chords in letter sequences, learn to use them in combination with single key presses
from 8	for all exercises: try to type without looking at the hands		increase performance by avoiding constant shift of visual attention
10	public performance	sentences requested by audience	free text generation and speed up
from 11	longer sentences		speed up
from 12	focus on slow letter sequences	<i>incredible complicated reservations somewhere</i>	speed up
from 13	play random letter sequences on a particular part of the piano	<i>edxpheth geedet heweth thad etamte</i>	practice hand combination on different parts of piano
	short texts	see Figure 4	speed up
from 14	practice words with high error rate		decrease error rate
	metronome: type sentences to a beat, successively increase beat	see Figure 5	speed up and decrease error rate
from 15	inner hearing: imagine the sound of a word before typing it		improve performance by making use of the audio feedback
	jump: practice words that require larger lateral arm movements		speed up arm movements and reduce the need for visual attention
from 17	sentences with many chords	<i>what would you do but what about her see Figure 6</i>	improve inter-chord interval
	German sentences	sentences from [3] translated into german such as: <i>ich werde dich morgen nacht anrufen wenn ich zu- rueck komme</i>	quickly resolve multiplicity for unpracticed letter sequences, spontaneously decide for best translation
from 22	practice hand-alternation before and after chord usage	<i>there was a lot going on be- tween now and then i would like to take care of you and your sister</i>	improve speed for chords

Table 1: The exercise program for learning fast text entry with PianoText.

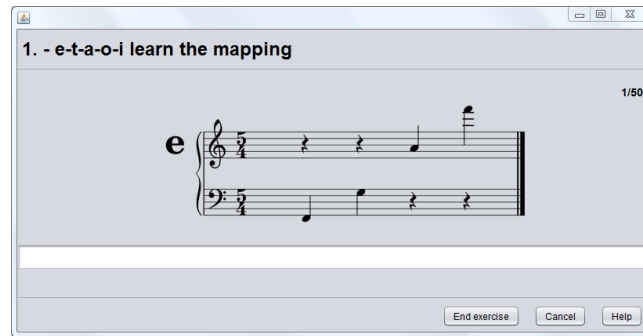


Figure 2: Very first exercise that trains the user to memorize the mapping. The notes assigned to each letter are presented and the user has to press each one of them.

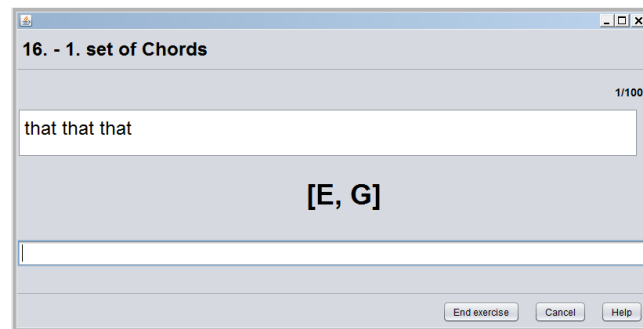


Figure 3: Exercise that introduces the chord mapping. The note names are shown in the bottom. The user has to enter the given n-gram three times by pressing the chord in different octaves.



Figure 4: In this exercise the user has to write a small text. This improves performance by practicing the spontaneous input of text. The texts are taken from [2]

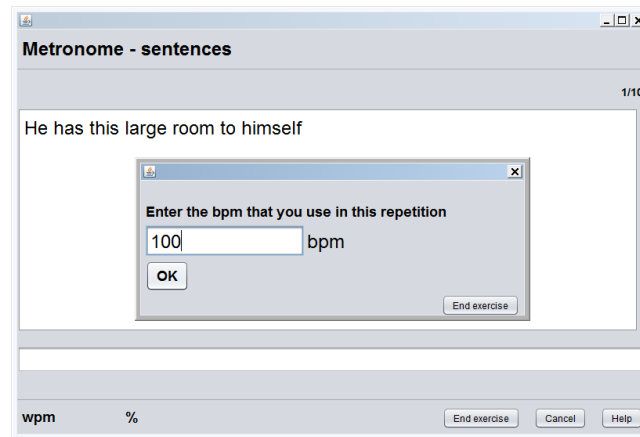


Figure 5: A metronome exercise. The user has to enter the beats per minutes. Regular tones then indicate the speed at which the keys have to be pressed.

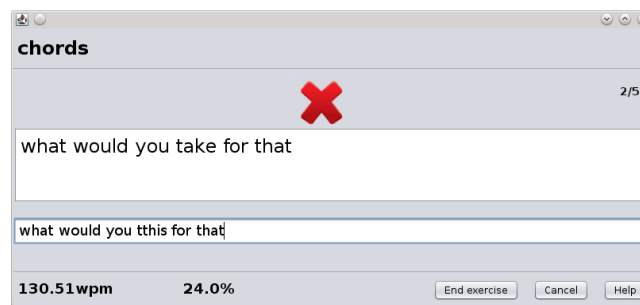


Figure 6: The sentence in this exercise contains 5 n-grams that can be entered by a chord: ‘what’, ‘would’, ‘take’ and ‘that’. The goal is to decrease the time between the key presses. The red cross is shown after entering the last word and indicates that the user made an error.

Exercise sentences from

- [1] Kelly, C. A website for studying english as a second language. www.manythings.org.
- [2] Lee, R. C. English as a second language. www.rong-chang.com.
- [3] Vertanen, K., and Kristensson, P. A versatile dataset for text entry evaluations based on genuine mobile emails. In *Proc. MobileHCT'11*, ACM (2011), 295–298.